

CS 411 Lab 2

Rogue-Like Algebra

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Version 2

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1 Introduction

Section 1 and the subsections therein provide the necessary contexts to understand the requirements of *Rogue-like Algebra* (e.g. definitions and references).

1.1 Purpose

This Software Requirements Specifications document will provide detailed descriptions for specifications and requirements for *Rogue-like Algebra*, a video game designed to aid in the learning of various algebraic concepts.

1.2 Scope

In our current world, students are finding it harder to learn and understand algebraic concepts through the traditional ways of teaching algebra. To try to solve this issue, a solution should address the issues that make the traditional means of teaching ineffective by being an interactive and more engaging medium of learning such concepts. *Rogue-like Algebra* aims to teach algebraic concepts that students find themselves struggle to absorb through engaging and fun gameplay in the rogue-like genre.

1.3 Definitions, Acronyms, and Abbreviations

- Software Requirements Specifications (SRS) – An outlining of a software’s purpose, requirements, and specifications. This acts as an overall guide to the development process of a software.
- Adaptive Learning – An educational method that adjusts content difficulty based on the learner’s performance to personalize the learning experience.
- Algebraic Concepts – Foundational mathematical principles involving variables, equations, expressions, and functions typically taught in middle and high school.
- Continuous Integration (CI) – A development practice where developers frequently merge code changes into a shared repository, triggering automated builds and tests to detect issues early.
- Continuous Deployment (CD) – A software release process that automatically delivers validated code changes to production, minimizing manual intervention and ensuring faster updates.
- GDScript – A Python-like scripting language used in the Godot game engine, designed for creating game logic.
- Git – A distributed version control system that tracks changes in source code during software development, enabling collaboration and history tracking.
- GitHub – A cloud-based platform for hosting Git repositories. It offers tools for collaboration, issue tracking, code review, and CI/CD integration.
- Godot – A free and open-source game engine used for developing 2D and 3D games, known for its flexibility and ease of use.
- Integrated Development Environment (IDE) – A software suite that provides tools like a code editor, debugger, and compiler to streamline software development (e.g., VS Code).
- Procedural Generation – A method of creating data algorithmically rather than manually, often used for generating levels, puzzles, or content dynamically in games.
- Rogue-like – A subgenre of games typically featuring procedural generation, turn-based movement, and permanent death, emphasizing strategy and replayability.
- Role-playing Game (RPG) – A genre of game in which a character assumes a role to complete various goals whether it be a larger story-line, or less connected quests, or both.
- Rust – A systems programming language known for performance, safety, and concurrency.
- Turn-Based Combat – A gameplay mechanic where players and opponents take alternating turns to perform actions such as attacking or defending.
- Visual Studio Code (VS Code) – A popular, lightweight code editor developed by Microsoft, used for writing and debugging code.

1.4 References

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1.5 Overview

Section 2 of this document will provide an overview of *Rogue-like Algebra*.

Section 3 of this document will provide a list of requirements that the product should fulfill.

2 Overall Description

Section 2 and the subsections contained within discuss *Rogue-like Algebra* in more detail with the overall perspective, functions, and those who will be using it.

2.1 Product Perspective

Rogue-like Algebra is a web-based application that is designed to be widely accessible on any web browser to help in the idea of being an effective way for people to learn and help build core algebraic principles. Featuring elements from various genres of video games including RPGs and rogue-likes, *Rogue-like Algebra* aims to add a twist of algebra to the gameplay loop many are familiar with to help those who struggle with learning such concepts.

2.2 Product Functions

Rogue-like Algebra will be a web-based application that will feature concepts that are common to the top-down RPG genre of video games including a character the player controls in different states of the game (e.g. overworld game state or battle game state) with enemy characters that behave according to the state of the game and reactive to the player character's actions. The various enemy characters will be able to engage the player in turn-style battle where procedural problems are created for the player to end the battle victoriously. The player will be able to encounter various equipment and power-ups in their gameplay whether it be from an active battle or the more passive exploration of the world.

2.3 User Characteristics

Rogue-like Algebra has three major groups of users who benefit the most from the educational goals the game aims to assist in: Students (or anybody seeking to improve their algebra skills, but students will be the biggest part of this group), those who can assist in first group's teachings, being teachers and parents, and the wider video game consuming audience. Students can use this software with the intent to either learn or strengthen knowledge of algebraic principles using the medium's higher engagement capabilities.

2.4 Constraints

N/A

2.5 Assumptions and Dependencies

N/A